

## EXP15: Decision Tree

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**Aim:**

To develop a Python program that implements a Decision Tree Classifier, so that the given dataset is used to build a tree model and predict the class label of a new sample.

**Algorithm:**

Step 1: Read the training dataset consisting of feature values and their corresponding class labels.

Step 2: Select the best attribute to split the dataset using a measure such as entropy or information gain.

Step 3: Split the dataset into subsets based on the values of the selected attribute.

Step 4: Check whether the resulting node is pure (all samples belong to one class) or a stopping condition is met.

Step 5: If the node is pure, create a leaf node with the corresponding class label.

Step 6: Otherwise, repeat Steps 2 to 5 recursively for each subset to build the tree further.

Step 7: Use the trained decision tree model to predict the class label of a new, unseen test sample.

Step 8: Display the structure of the decision tree and the predicted output.

**Source Code:**

```
# Program to implement Decision Tree
```

```
from sklearn.tree import DecisionTreeClassifier
from sklearn import tree
```

```
# Dataset: Weather conditions to decide whether to Play Tennis
```

```
# Features: Outlook(0=Sunny,1=Overcast,2=Rain), Temperature(0=Hot,1=Mild,2=Cool),
Humidity(0=High,1=Normal), Wind(0=Weak,1=Strong)
```

```
X = [
    [0, 0, 0, 0],
    [0, 0, 0, 1],
    [1, 0, 0, 0],
    [2, 1, 0, 0],
    [2, 2, 1, 0],
    [2, 2, 1, 1],
    [1, 2, 1, 1],
    [0, 1, 0, 0],
    [0, 2, 1, 0],
    [2, 1, 1, 0],
]
```

```
y = ['No', 'No', 'Yes', 'Yes', 'Yes', 'No', 'Yes', 'No', 'Yes', 'Yes']
```

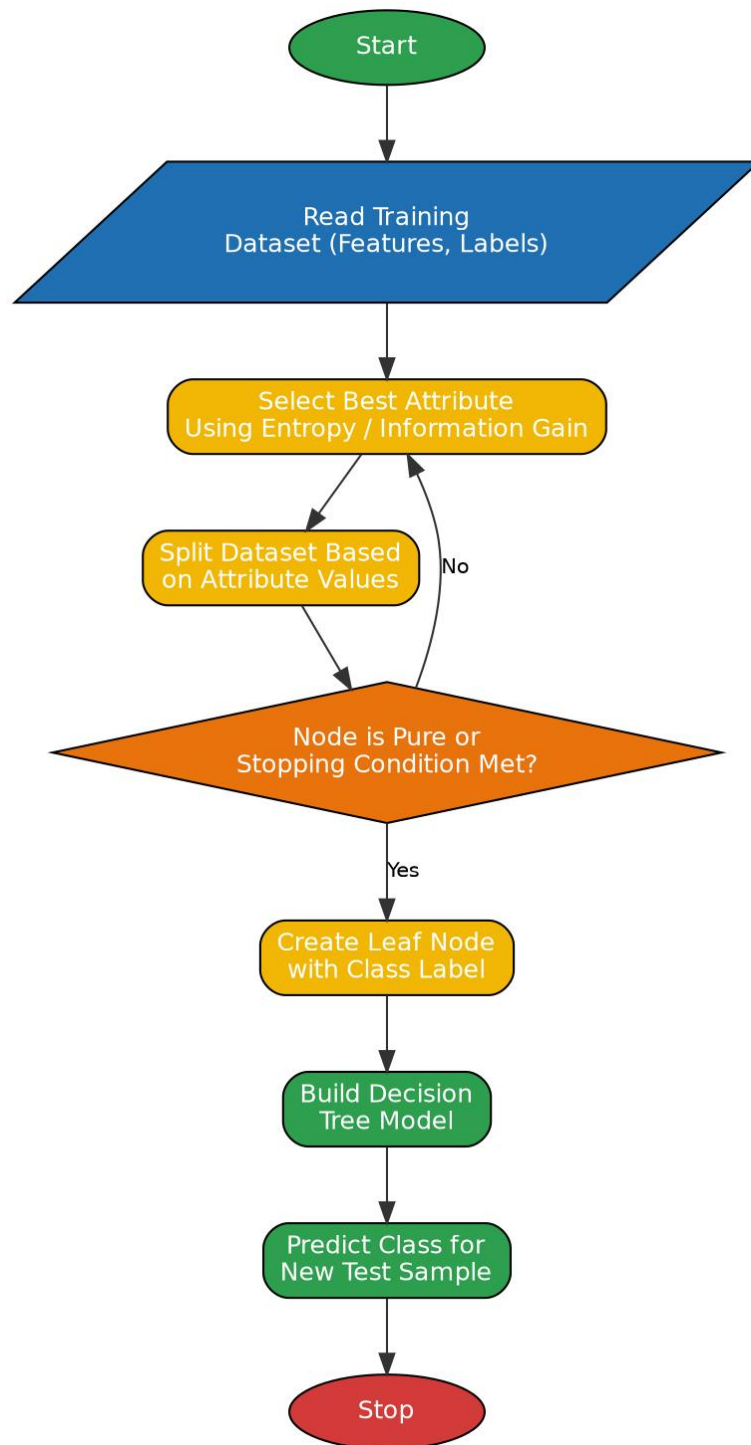
```
model = DecisionTreeClassifier(criterion='entropy', random_state=0)
```

```
model.fit(X, y)

feature_names = ['Outlook', 'Temperature', 'Humidity', 'Wind']
tree_rules = tree.export_text(model, feature_names=feature_names)
print("Decision Tree Structure:")
print(tree_rules)

test_sample = [[0, 1, 0, 1]] # Sunny, Mild, High Humidity, Strong Wind
prediction = model.predict(test_sample)
print("Prediction for sample [Sunny, Mild, High, Strong]:", prediction[0])
```

**Flowchart:**



## Output:

```
>>> ===== RESTART: C:/Users/yanar/Documents/CSA1717/EXP15.py =====
Decision Tree Structure:
|--- Outlook <= 0.50
|   |--- Temperature <= 1.50
|   |   |--- class: No
|   |--- Temperature > 1.50
|   |   |--- class: Yes
|--- Outlook > 0.50
|   |--- Wind <= 0.50
|   |   |--- class: Yes
|   |--- Wind > 0.50
|   |   |--- Outlook <= 1.50
|   |   |   |--- class: Yes
|   |   |--- Outlook > 1.50
|   |   |   |--- class: No
Prediction for sample [Sunny, Mild, High, Strong]: No
>>> |
```

## Result:

Thus the program to implement the Decision Tree was written, executed successfully and the output was verified.